

## Revealing a human p53 universe

Thuy-Ai T. Nguyen<sup>1\*</sup>, Sara A. Grimm<sup>2</sup>, Pierre R. Bushel<sup>3</sup>, Jianying Li<sup>2</sup>, Yuanyuan Li<sup>3</sup>, Brian D. Bennett<sup>2</sup>, Christopher A. Lavender<sup>2</sup>, James M. Ward<sup>2</sup>, David C. Fargo<sup>2, 4</sup>, Carl W. Anderson<sup>1</sup>, Leping Li<sup>3</sup>, Michael A. Resnick<sup>1\*</sup>, and Daniel Menendez<sup>1\*</sup>

<sup>1</sup>Genome Integrity & Structural Biology Laboratory, National Institute of Environmental Health Sciences/National Institutes of Health, Research Triangle Park, North Carolina, 27709, USA

<sup>2</sup>Integrative Bioinformatics Support Group, National Institute of Environmental Health Sciences/National Institutes of Health, Research Triangle Park, North Carolina, 27709, USA

<sup>3</sup>Biostatistics & Computational Biology Branch, National Institute of Environmental Health Sciences/National Institutes of Health, Research Triangle Park, North Carolina, 27709, USA

<sup>4</sup>Office of Scientific Computing, National Institute of Environmental Health Sciences/National Institutes of Health, Research Triangle Park, North Carolina, 27709, USA

### **\*Joint corresponding authors**

\*Corresponding authors: Thuy-Ai T. Nguyen, Tel: 1-919-541-4821; Email: [thuy-ai.nguyen@nih.gov](mailto:thuy-ai.nguyen@nih.gov). Correspondence may also be addressed to Daniel Menendez, Tel: 1-919-541-0972; Email: [menendez@niehs.nih.gov](mailto:menendez@niehs.nih.gov); or Michael A. Resnick, Tel: 1-919- 541-4480; Email: [resnick@niehs.nih.gov](mailto:resnick@niehs.nih.gov).

## **Supplementary Materials and Methods**

### **Lymphocyte isolation**

Healthy adult volunteers were recruited to the NIEHS Clinical Research Unit and underwent phlebotomy per the approved NIEHS institutional review board (IRB#07-E-0023) protocol. All volunteers gave written informed consent for blood donation. Volunteers were excluded if they had a history of recent infection, were on anti-inflammatory medications, or tested positive for hepatitis B, C or HIV. Up to 320 ml of whole blood were withdrawn from an antecubital vein into citrated tubes. Human lymphocytes were isolated with Histopaque-1077 (Sigma-Aldrich) per manufacturer's protocol from the blood of four healthy volunteers. T- lymphocytes were activated with 1.5% vol/vol phytohemagglutinin-M (PHA, Invitrogen) and grown in RPMI supplemented with 10% FBS (Fetal Bovine Serum, Invitrogen, Carlsbad, CA) and maintained at 37°C with 5% CO<sub>2</sub> for 48h prior to drug treatment.

### **ChIP-seq sample preparation and sequencing**

PHA-stimulated T lymphocytes were treated with 0.3 µg/ml Doxorubicin, 10 µM Nutlin-3, or 0.1% DMSO. Cells were harvested 72h post PHA stimulation and 24h post drug treatment. 17-30 million cells were used per each ChIP-seq experiment. Chromatin was sheared to 100-300 bp using a Bioruptor sonicator (Diagenode) for 60 minutes at high power, 30 seconds ON, 30 seconds OFF. ChIP assays were performed as described in our previous work (Menendez, NAR 2013). DNA was quantified using a Qubit Fluorometer (Invitrogen). Libraries were prepared using Illumina's ChIP-seq Sample Prep kit per manufacturer's instructions and were sequenced by the National Center for Genome Resources (NCGR, Santa Fe, NM) on the Illumina Genome Analyzer GAIIx to yield 36-bp reads (Illumina, Inc., San Diego, CA). Sequencing data for these data sets have been deposited in GEO under accession GSE110368.

### **Lymphocyte RNA and microarray sample preparation**

PHA-stimulated T lymphocytes were treated with 0.3 µg/ml Doxorubicin, 10 µM Nutlin-3, or 0.1% DMSO. Cells were harvested 72h post PHA stimulation and 24h post drug treatment. 3-4 million cells were used per each replicate sample. Triplicate RNA samples for each volunteer were isolated using the RNeasy Mini Kit (Qiagen, Valencia, CA, USA) with DNase treatment per the manufacturer's protocol. Microarrays were performed in the Affymetrix GeneChip Instrument system using procedures as previously described (Menendez, NAR 2013). The RNA was prepared for microarray expression analysis by the NIEHS Microarray Core using the GeneChip 3' IVT Express kit protocol as per the manufacturer's instructions and the Affymetrix Human Genome U133 Plus 2.0 GeneChip® arrays (Affymetrix, Santa Clara, CA). Arrays were scanned in an Affymetrix Scanner 3000, and data were obtained using the GeneChip® Command Console Software (AGCC; Version 1.1). Microarray data for these data sets have been deposited in GEO under accession GSE110369.

### **Statistical Analysis of G4 colocalization with p53 peaks**

The statistical significance of the colocalization of p53 peaks with G4 regions was determined by Monte Carlo simulation, where BEDtools v2.24.0 'shuffle' was used to select random size-

matched genomic regions (excluding non-canonical chromosomes, ENCODE blacklisted regions, and regions with CRG 36mer alignability score less than 0.5) for 10,000 iterations.

### **Word cloud plot**

To reveal the important and frequently appearing cistrome genes from the 943 p53 cistrome genes, we utilized a text-mining strategy. All the “word cloud” plots were done in R (<https://www.r-project.org/>) with its packages (Fellows 2014, Lang 2016, Feinerer and Hornik 2017). In the plots, the larger the fonts, the more frequent the genes appeared.

### **Supplementary materials and methods references:**

Feinerer, I. and K. Hornik (2017). "tm: Text Mining Package." (<https://cran.r-project.org/web/packages/tm/index.html>)

Fellows, I. (2014). "wordcloud: Word Clouds." (<https://cran.r-project.org/web/packages/wordcloud/index.html>)

Lang, D. (2016). "wordcloud2: Create Word Cloud by htmlWidget." (<https://cran.r-project.org/web/packages/wordcloud2/index.html>)

Menendez, D., Nguyen, T.A., Freudenberg, J.M., Mathew, V.J., Anderson, C.W., Jothi, R., and Resnick, M.A. (2013). Diverse stresses dramatically alter genome-wide p53 binding and transactivation landscape in human cancer cells. *Nucleic Acids Res* 41, 7286-7301.

## **Supplementary Table Legends**

**Supplementary Table ST1.** Data sets used for meta-analysis. The excel contains the references, cell lines, treatment conditions, inputs, published & SISR's total peaks called, and expression data corresponding to the studies that were analyzed in this paper.

**Supplementary Table ST2.** All 54,947 activated p53 peaks

**Supplementary Table ST3.** All 16,209 control (NT/DMSO) p53 peaks

**Supplementary Table ST4.** The relationship between perfect motifs with 0–15 basespacers, or perfect 20-mer with 1 mismatch, and in vivo p53 binding

**Supplementary Table ST5.** p53 peaks in  $\geq 2$  data sets and their association with miRNA, lncRNA, eRNA, or G4

**Supplementary Table ST6.** All 7,880 p53-associated transcriptome DEGs

**Supplementary Table ST7.** Cistrome analysis (16,882 p53 peaks nearTSS and expression status)

**Supplementary Table ST8.** p53 cistrome targets

1. Tab 8a: 943 Direct targets with one p53 bound peak per cistrome target gene
2. Tab 8b: All 1,840 activated p53 peaks nearTSS with associated gene expression

**Supplementary Table ST9:** Comparison of DXR vs. Nutlin-3 binding, transcriptome & cistrome

## Supplementary Figures Legends

**Supplementary Figure SF1. Analysis workflow of p53 data sets for binding, transcriptome and cistrome analysis.** Dashed red arrows indicate data sets that were excluded from the analysis.

**Supplementary Figure SF2. The human p53 binding and expression resource (BAER) hub.** A UCSC genome browser screen shot of *CDKN1A* in the human p53 BAER hub with a description of the customizable views.

**Supplementary Figure SF3. Genomic distribution of peaks relative to genic regions from the 41 activated p53 ChIP-seq data sets divided into cancer, normal of p53 overexpressing cells.** Presented are 20 data sets in the cancer category (LCL, CAL51, HCT116, MCF7, U2OS); 19 data sets in the normal category (BJ, FSF, HFK, IMR90, Lymph, PBMC, hESC); and 2 data sets in the p53 overexpressing category (SAOS2). TSS, 5 kb upstream through the first intron; intergenic, beyond 3'UTR up to 5kb upstream of the next TSS. Each point is a ChIP-seq data set. Tukey box plot: box = 25th-75th percentile, middle line = median, and whiskers = 1.5 \* interquartile range (IQR).

**Supplementary Figure SF4. Distribution of p53 peaks across human chromosome 5 from all 41 data sets compared to protein coding genes and CpG islands.** Data set reference citations are in ( ) following the peak descriptions.

**Supplementary Figure SF5: Distribution of p53 motif categories in individual data sets.** Tukey boxplot of peaks from the 44 activated p53 ChIP-seq data sets based on the type of p53 motif bound: 0-spacer = p53 consensus motif containing two decamers with no spacer sequence (mean = 45%), spacers = with spacers of 1-15 bp between the decamers (mean = 13%), half-site = one decamer (mean = 1%). Each point is a data set. Tukey box plot: box = 25th-75th percentile, black line = median, and whiskers = 1.5 \* interquartile range (IQR). Open symbols are outliers.

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**Supplementary Figure SF7. Dynamic analysis of p53 ChIP-seq data by ORIO.** ORIO (Online Resource for Integrative Omics) (Lavender et al., 2017) was used to generate a dynamic webpage describing trends in p53 ChIP-seq data. Hierarchical clustering was performed considering the Spearman correlation of input-subtracted read coverage signal across p53 peaks that appear in  $\geq 20$  data sets. Clustering results are displayed in a heatmap; the color of individual cells describes the correlation value between two datasets and the cluster distances

between each sample shown using a dendrogram (left). This and other views, such as plots describing the correlation between each dataset pair, can be found at <https://www.niehs.nih.gov/research/resources/databases/p53/index.cfm>.

**Supplementary Figure SF8: Enriched known motif sequences found by HOMER within p53 peaks in  $\geq 2$  data sets that lack a p53 motif.**

**Supplementary Figure SF9: Ingenuity Pathway analysis (IPA) of 2110 genes with a p53-like motif-containing peak in  $\geq 2$  data sets near the TSS (5 kb upstream through the first intron) reveals p53 involvement in novel biological processes and new transcriptional targets.** 25 genes identified as “p53 signaling genes” were removed and only IPA pathway categories with a Benjamini-Hochberg (BH) Multiple Testing Correction p-value  $< 1 \times 10^{-4}$  are reported.

**Supplementary Figure SF10: p53 binding peaks in  $\geq 20$  data sets are associated with enhancer regions.** Enrichment of p53 binding peaks that appears in  $\geq 20$  data sets in ENCODE HMM (hidden Markov Model) marks compared to p53 binding peaks that appear in only two data sets.

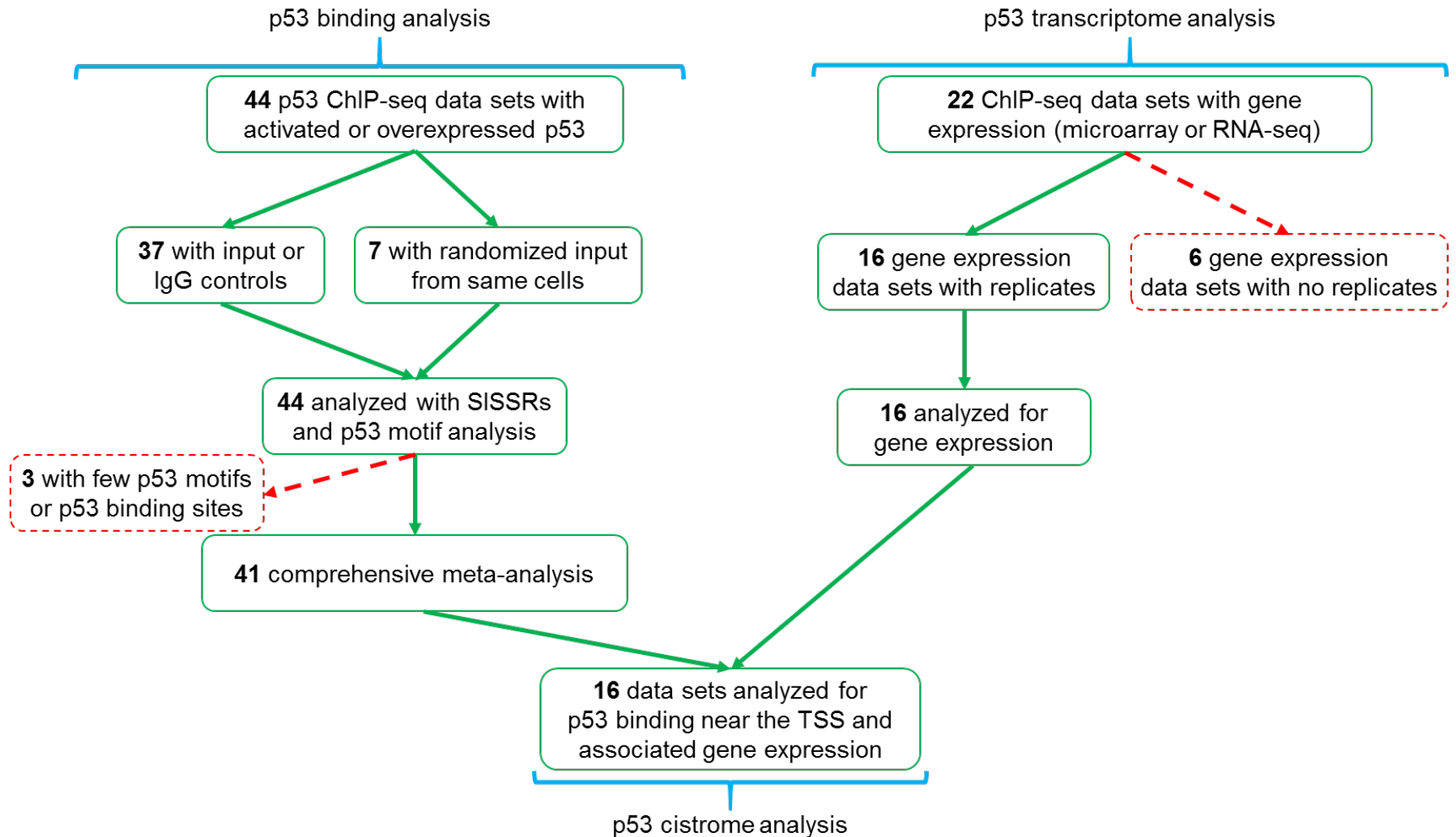
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**Supplementary Figure SF12: p53 transcriptome and p53 cistrome targets overlap with DREAM targets.** (A) p53-associated transcriptome targets in  $\geq 2$  data sets overlap substantially with both DREAM targets and p53 known targets; (B) p53 cistrome targets minimally overlap with DREAM targets as compared to p53 known targets.

**Supplementary Figure SF13: Word cloud images of p53 cistromes that are common in at least three data sets** after (A) DXR, (B) Nutlin, or (C) 24hr after treatment with various agents; or in (D) normal or (E) cancer cells. The size of each word indicates its frequency of overlap across data sets.

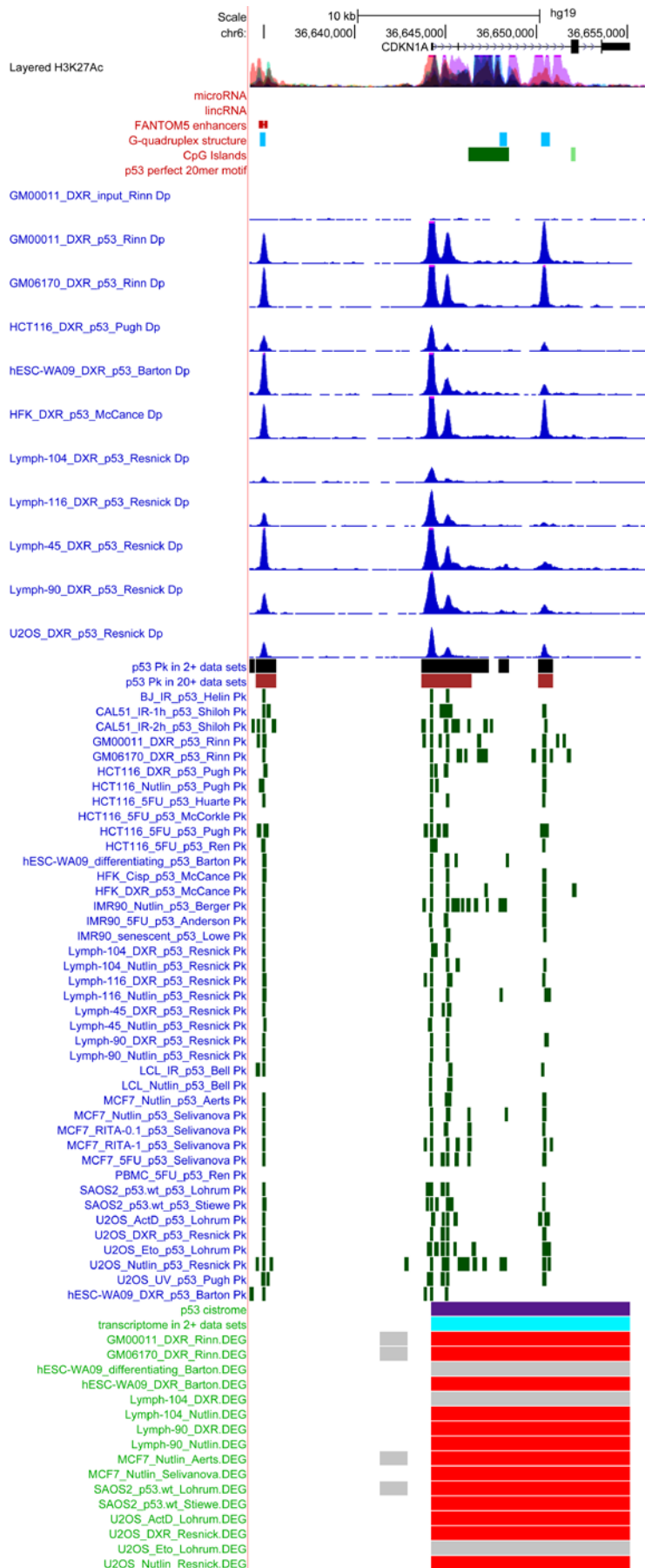
**Supplementary Figure SF14: A Venn diagram comparing the different p53 cistromes (DXR, Nutlin, 24hr after treatment with various agents, normal cells, cancer cells) derived from the present meta-analysis.** The data sets used in each category are not mutually exclusive.

**Supplementary Figure SF1. Analysis workflow of p53 data sets for binding, transcriptome and cistrome analysis.** Dashed red arrows indicate data sets that were excluded from the analysis.



# Supplementary Figure SF2. The human p53 binding and expression resource (BAER) hub

<https://orio.niehs.nih.gov/ucscview/nguyen/p53BAER/hub.txt>



p53 BAER hub with customizable views of:

## •Activated p53 ChIP-seq

– coverage track depth

■ SISSRs peak calls

■ p53 peaks in 2+ data sets

■ p53 peaks in 20+ data sets

## •Control p53 ChIP-seq

– coverage track depth

■ SISSRs peak calls

## •Input ChIP-seq coverage track depth

## •Other expression regulatory nodes

■ enhancers

■ lincRNA

■ microRNA

■ G-quadruplex structures

■ CpG Islands

■ p53 perfect 20mer motif

## •Differential gene expression associated with p53 ChIP-seq data sets

■ p53 cistrome target genes

■ transcriptome in 2+ data sets

■ upregulated DEG

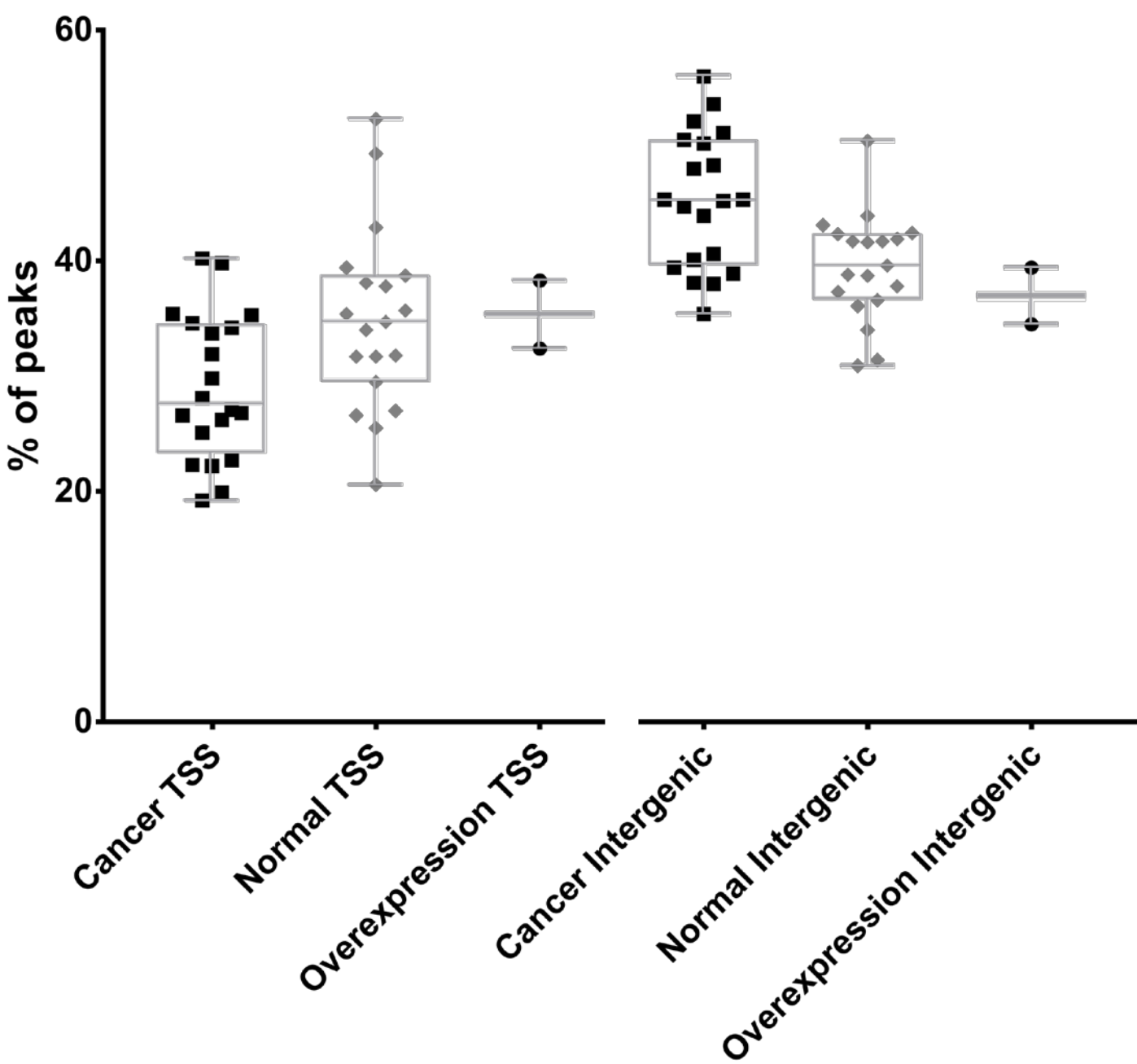
■ downregulated DEG

■ no change in expression

A UCSC genome browser screen shot of *CDKN1A* in the human p53 BAER hub with a description of the customizable views.



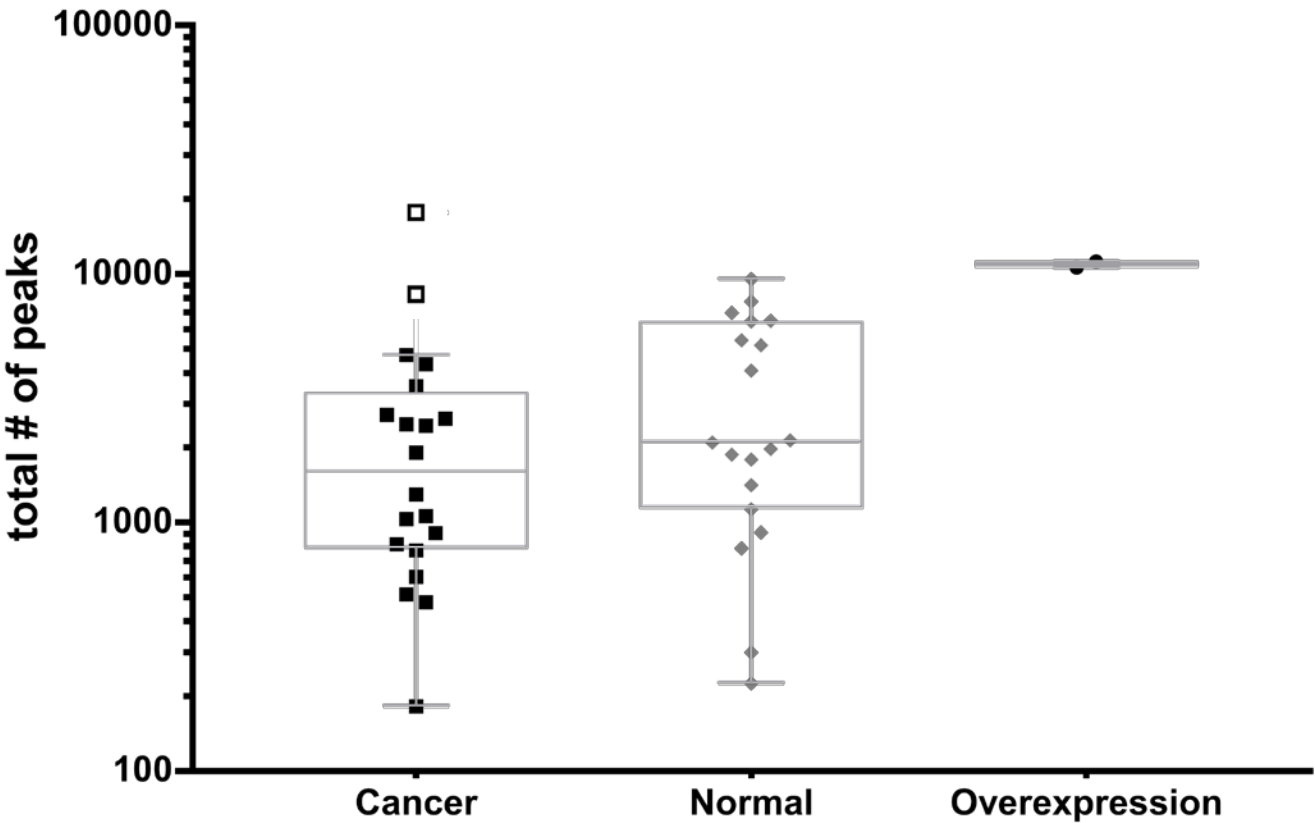
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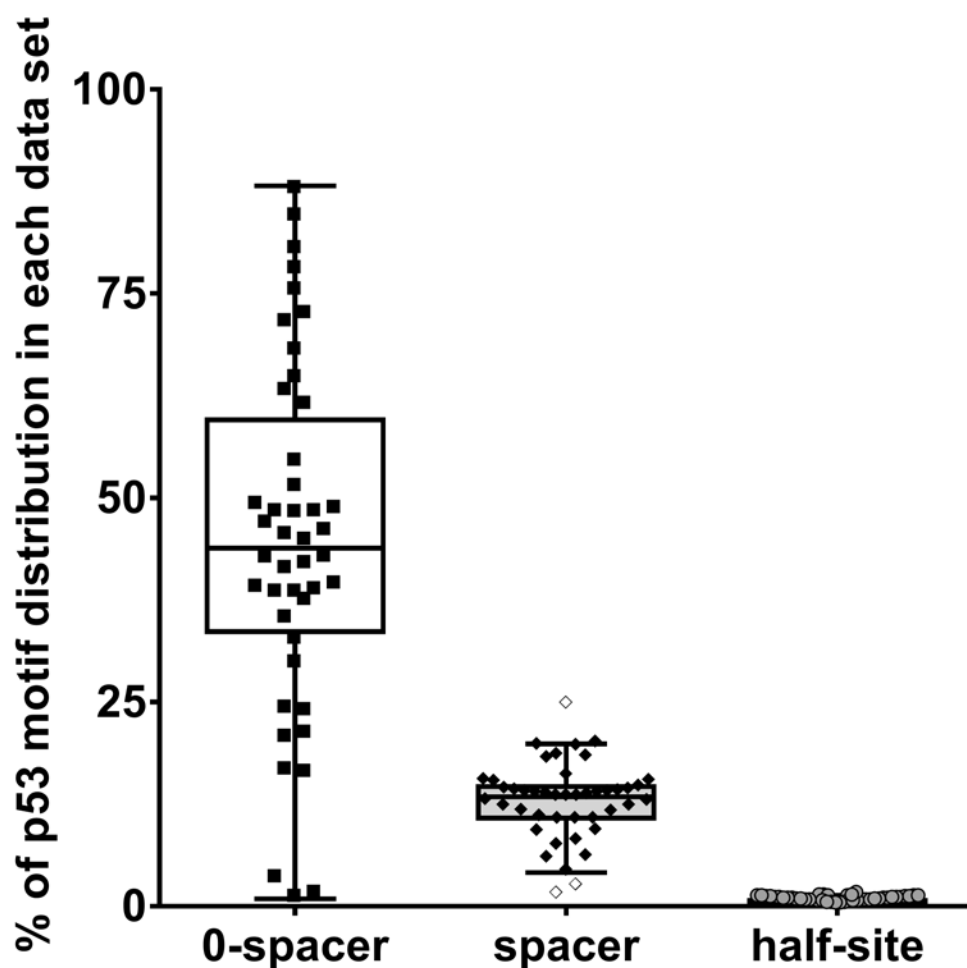
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**Supplemental Figure SF8: Enriched known motif sequences found by HOMER within p53 peaks in  $\geq 2$  data sets that lack a p53 motif.**

| <i>Logo</i>                                                                         | <i>TF family,<br/>representative motif<br/>and p-value</i> | <i>Related Motifs Among Top Hits</i>                                     |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|--------------------------------------------------------------------------|
|     | <b>p53 family</b><br>p53            1e-893                 | p63, p73, p53-myc                                                        |
|    | <b>bZIP family</b><br>fra2            1e-861               | fra1, junb, fosl2, atf3, batf,<br>jun-ap1, ap1, bach2, nfe2, bach1, nrf2 |
|     | <b>CTCF family</b><br>ctcf            1e-282               | boris                                                                    |
|   | <b>ETS family</b><br>fli1            1e-182                | elk1, etv2, elk4                                                         |
|  | <b>ETS family</b><br>ets1            1e-173                | gabp, erg, etv1, elf1, ets, ews-fli1                                     |
|  | <b>Runx family</b><br>runx1m        1e-126                 | runx1                                                                    |

**Supplemental Figure SF9: Ingenuity Pathway analysis (IPA) of 2110 genes with a p53-like motif-containing peak in  $\geq 2$  data sets peak near the TSS (5 kb upstream through the first intron) reveals p53 involvement in novel biological processes and new transcriptional targets.** 25 genes identified as “p53 signaling genes” were removed and only IPA pathway categories with a Benjamini-Hochberg (BH) Multiple Testing Correction p-value  $< 1 \times 10^{-4}$  are reported.

*Signaling Pathway Categories:*

*G-Protein Coupled Receptor Signaling*

RhoGDI Signaling (ratio: 39/173; BH p-value =  $1.6 \times 10^{-6}$ )  
Signaling by Rho Family GTPases (ratio: 44/234; BH p-value =  $1.5 \times 10^{-5}$ )  
G-Protein Coupled Receptor Signaling (ratio: 45/256; BH p-value =  $5.3 \times 10^{-5}$ )  
G Beta Gamma Signaling (ratio: 22/88; BH p-value =  $7.1 \times 10^{-5}$ )  
Gαq Signaling (ratio: 28/147; BH p-value =  $5.3 \times 10^{-4}$ )  
Phospholipase C Signaling (ratio: 39/237; BH p-value =  $5.5 \times 10^{-4}$ )

*Neurotransmitters and Other Nervous System Signaling*

Ephrin Receptor Signaling (ratio: 37/174; BH p-value =  $8.3 \times 10^{-6}$ )  
Cholecystokinin/Gastrin-mediated Signaling (ratio: 26/101; BH p-value =  $1.0 \times 10^{-5}$ )  
Axonal Guidance Signaling (ratio: 69/434; BH p-value =  $1.1 \times 10^{-5}$ )  
Glioma Signaling (ratio: 24/98; BH p-value =  $4.8 \times 10^{-5}$ )  
Glioblastoma Multiforme Signaling (ratio: 29/146; BH p-value =  $2.2 \times 10^{-4}$ )  
CREB Signaling in Neurons (ratio: 31/171; BH p-value =  $5.3 \times 10^{-4}$ )  
GNRH Signaling (ratio: 25/129; BH p-value =  $8.2 \times 10^{-4}$ )

*Growth Factor Signaling*

HGF Signaling (ratio: 25/105; BH p-value =  $4.8 \times 10^{-5}$ )  
ErbB Signaling (ratio: 21/86; BH p-value =  $1.5 \times 10^{-4}$ )  
NGF Signaling (ratio: 24/107; BH p-value =  $1.6 \times 10^{-4}$ )  
Neuregulin Signaling (ratio: 21/88; BH p-value =  $2.0 \times 10^{-4}$ )  
Neurotrophin/TRK Signaling (ratio: 17/67; BH p-value =  $4.8 \times 10^{-4}$ )  
PDGF Signaling (ratio: 18/77; BH p-value =  $7.2 \times 10^{-4}$ )

*Cellular Immune Response*

Leukocyte Extravasation Signaling (ratio: 41/198; BH p-value =  $6.7 \times 10^{-6}$ )  
Tec Kinase Signaling (ratio: 32/157; BH p-value =  $6.3 \times 10^{-5}$ )  
Virus Entry via Endocytic Pathways (ratio: 21/89; BH p-value =  $2.2 \times 10^{-4}$ )  
Role of Osteoblasts, Osteoclasts and Chondrocytes in Rheumatoid Arthritis (ratio: 38/219; BH p-value =  $2.7 \times 10^{-4}$ )

*Cardiovascular Signaling*

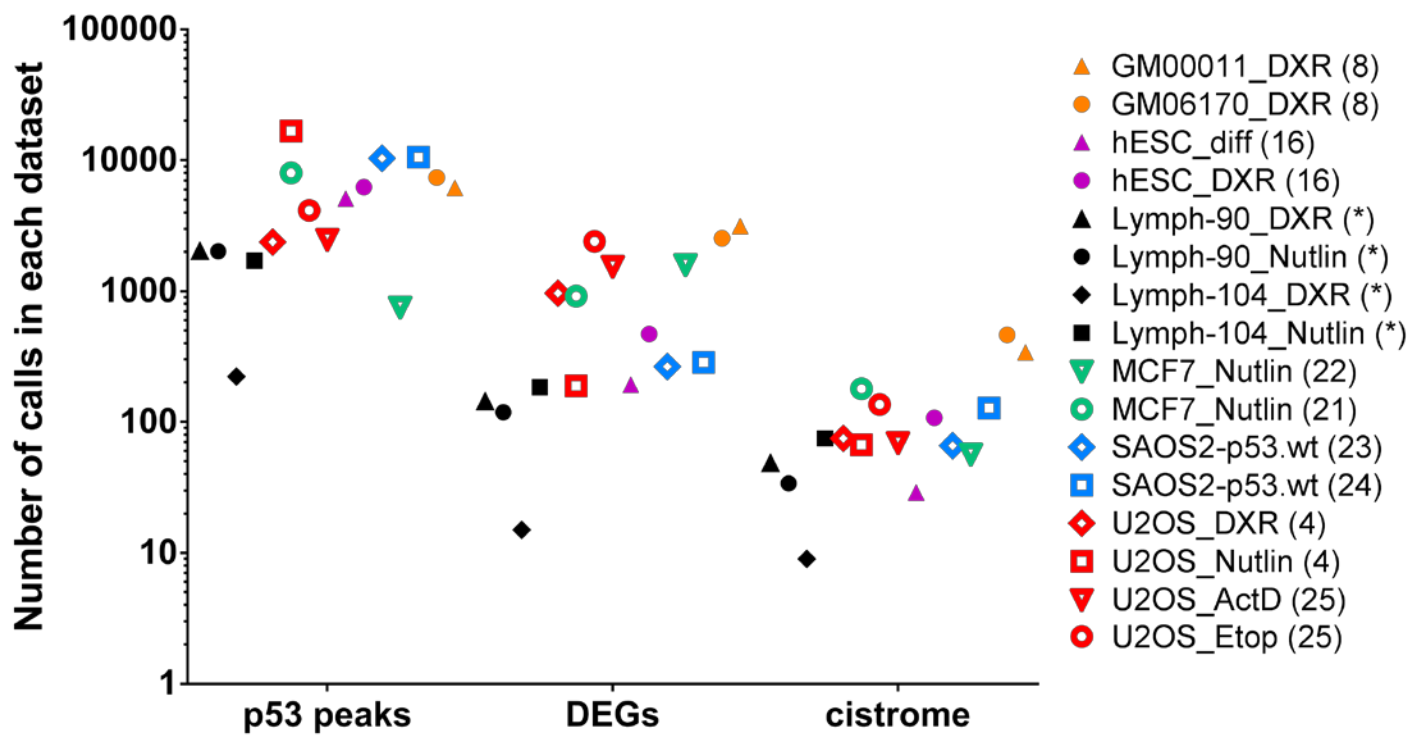
Role of NFAT in Cardiac Hypertrophy (ratio: 32/179; BH p-value =  $5.3 \times 10^{-4}$ )  
Thrombin Signaling (ratio: 33/190; BH p-value =  $6.9 \times 10^{-4}$ )

**Supplemental Figure SF10: p53 binding peaks in  $\geq 20$  data sets are associated with enhancer regions.** Enrichment of p53 binding peaks that appears in  $\geq 20$  data sets in ENCODE HMM (hidden Markov Model) marks compared to p53 binding peaks that appear in only two data sets.

| Cell line | Chromatin mark    | Enrichment <i>p</i> -value |
|-----------|-------------------|----------------------------|
| Gm12878   | 7_Weak_Enhancer   | 1.062-19                   |
|           | 5_Strong_Enhancer | 8.492-14                   |
|           | 4_Strong_Enhancer | 8.206-10                   |
| H1hesc    | 5_Strong_Enhancer | 7.441-19                   |
|           | 7_Weak_Enhancer   | 3.824-14                   |
|           | 4_Strong_Enhancer | 0.0072                     |
| Hepg2     | 7_Weak_Enhancer   | 9.480-13                   |
|           | 5_Strong_Enhancer | 1.823-10                   |
|           | 4_Strong_Enhancer | 0.0003                     |
| Hmec      | 5_Strong_Enhancer | 3.136-21                   |
|           | 4_Strong_Enhancer | 0.0155                     |
|           | 7_Weak_Enhancer   | 0.1197                     |
| Hsmm      | 5_Strong_Enhancer | 2.592-05                   |
|           | 7_Weak_Enhancer   | 0.0013                     |
|           | 4_Strong_Enhancer | 0.0017                     |
| Huvec     | 5_Strong_Enhancer | 4.631-07                   |
|           | 7_Weak_Enhancer   | 0.0012                     |
|           | 4_Strong_Enhancer | 0.0063                     |
| K562      | 7_Weak_Enhancer   | 0.0265                     |
|           | 4_Strong_Enhancer | 0.0318                     |
|           | 6_Weak_Enhancer   | 0.0357                     |
| Nhek      | 7_Weak_Enhancer   | 5.869-08                   |
|           | 5_Strong_Enhancer | 6.633-06                   |
|           | 4_Strong_Enhancer | 0.0055                     |
| Nhlf      | 5_Strong_Enhancer | 0.0013                     |
|           | 1_Active_Promoter | 0.0029                     |
|           | 7_Weak_Enhancer   | 0.0199                     |

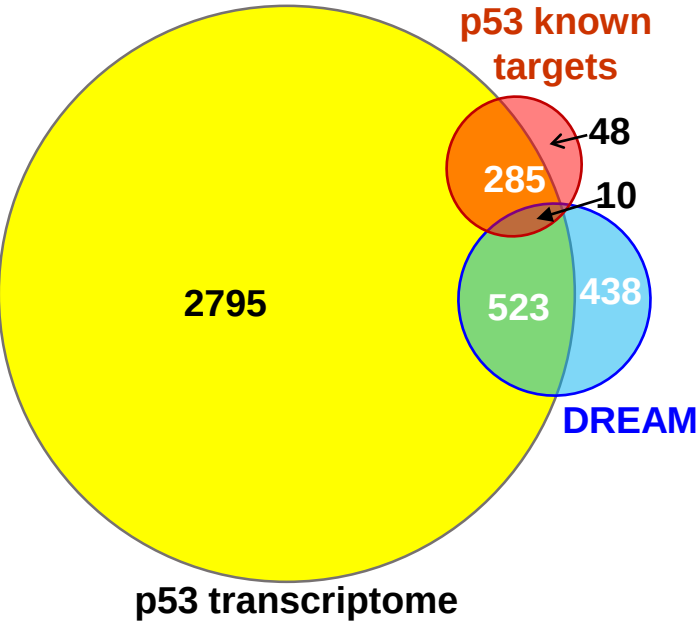


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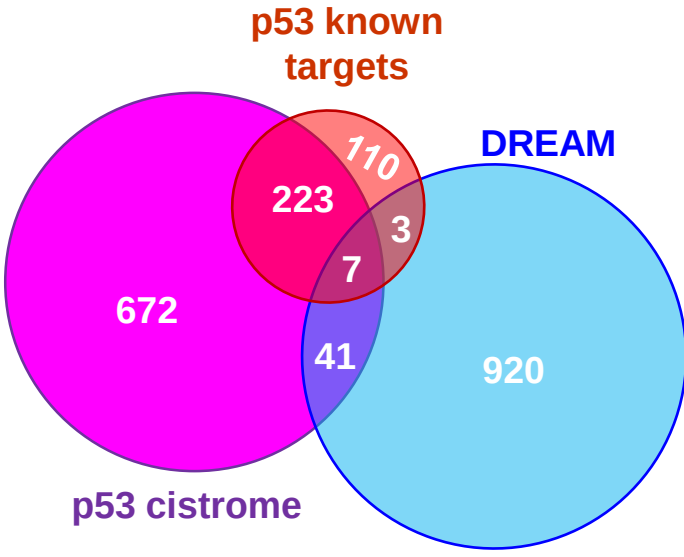


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SF12A



SF12B

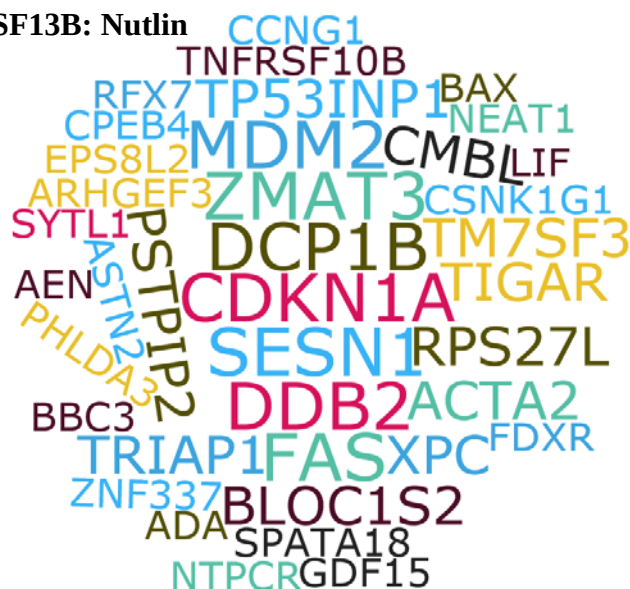


**Supplemental Figure SF13: Word cloud images of p53 cistrome genes that are common in at least three data sets after (A) DXR, (B) Nutlin, or (C) 24hr after treatment with various agents; or in (D) normal or (E) cancer cells. The size of each word indicates its frequency of overlap across data sets.**

### SF13A: DXR



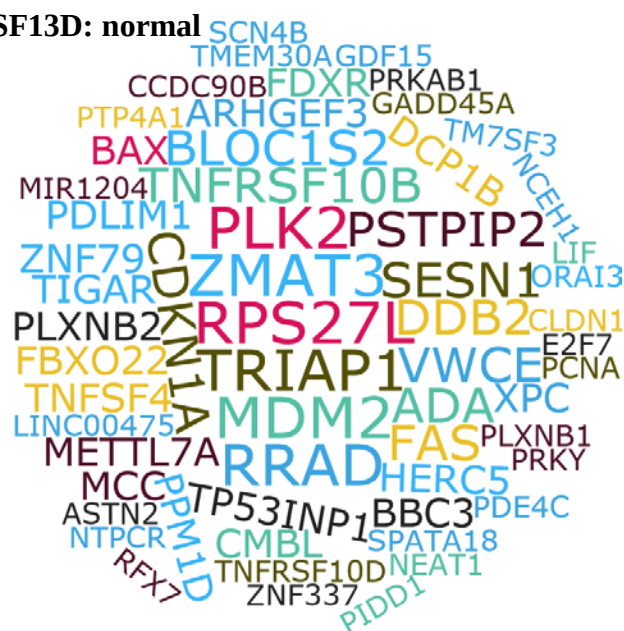
### SF13B: Nutlin



### SF13C: 24hr



### SF13D: normal



### SF13E: cancer

